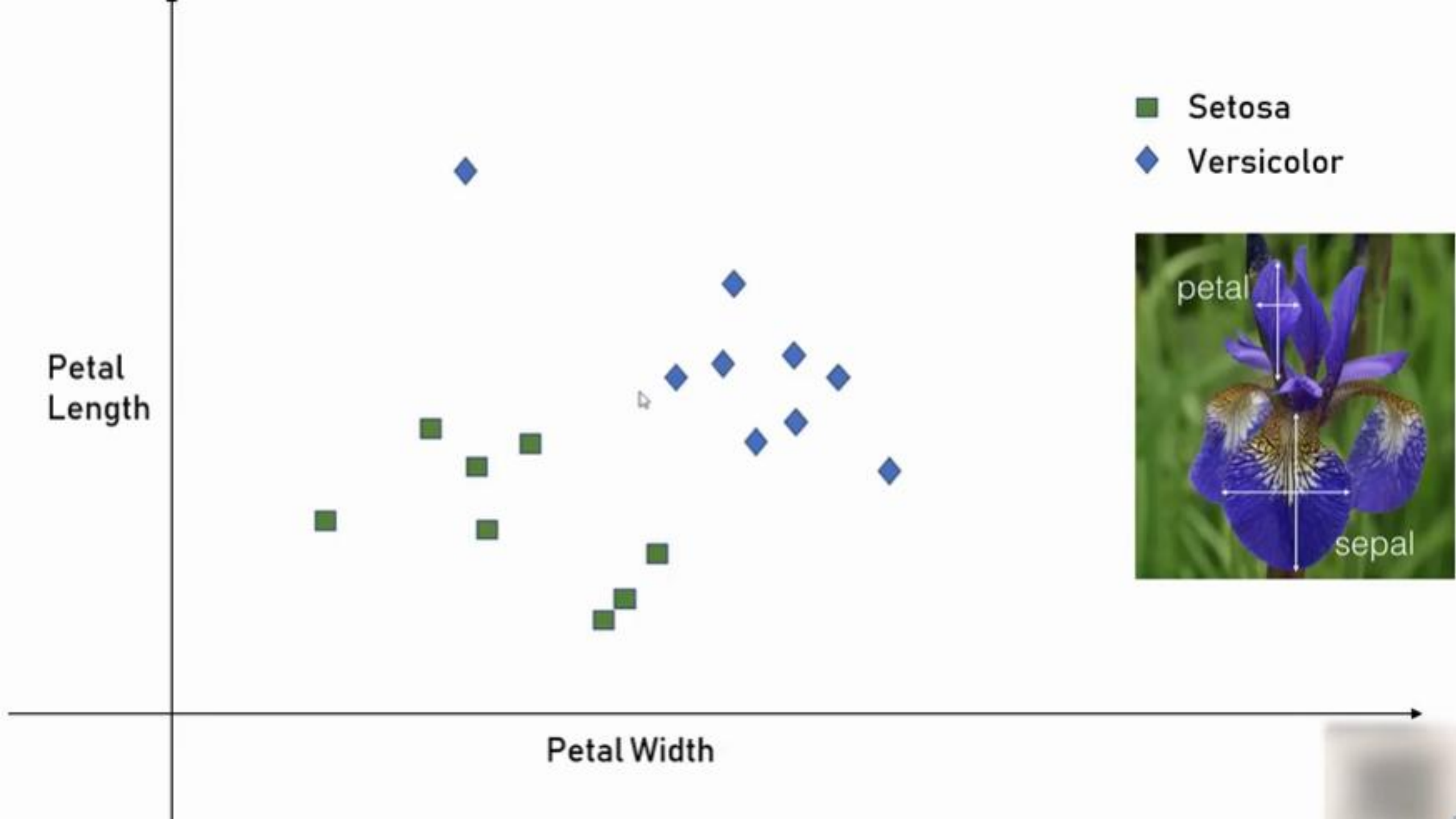
Advanced Classification Algorithms

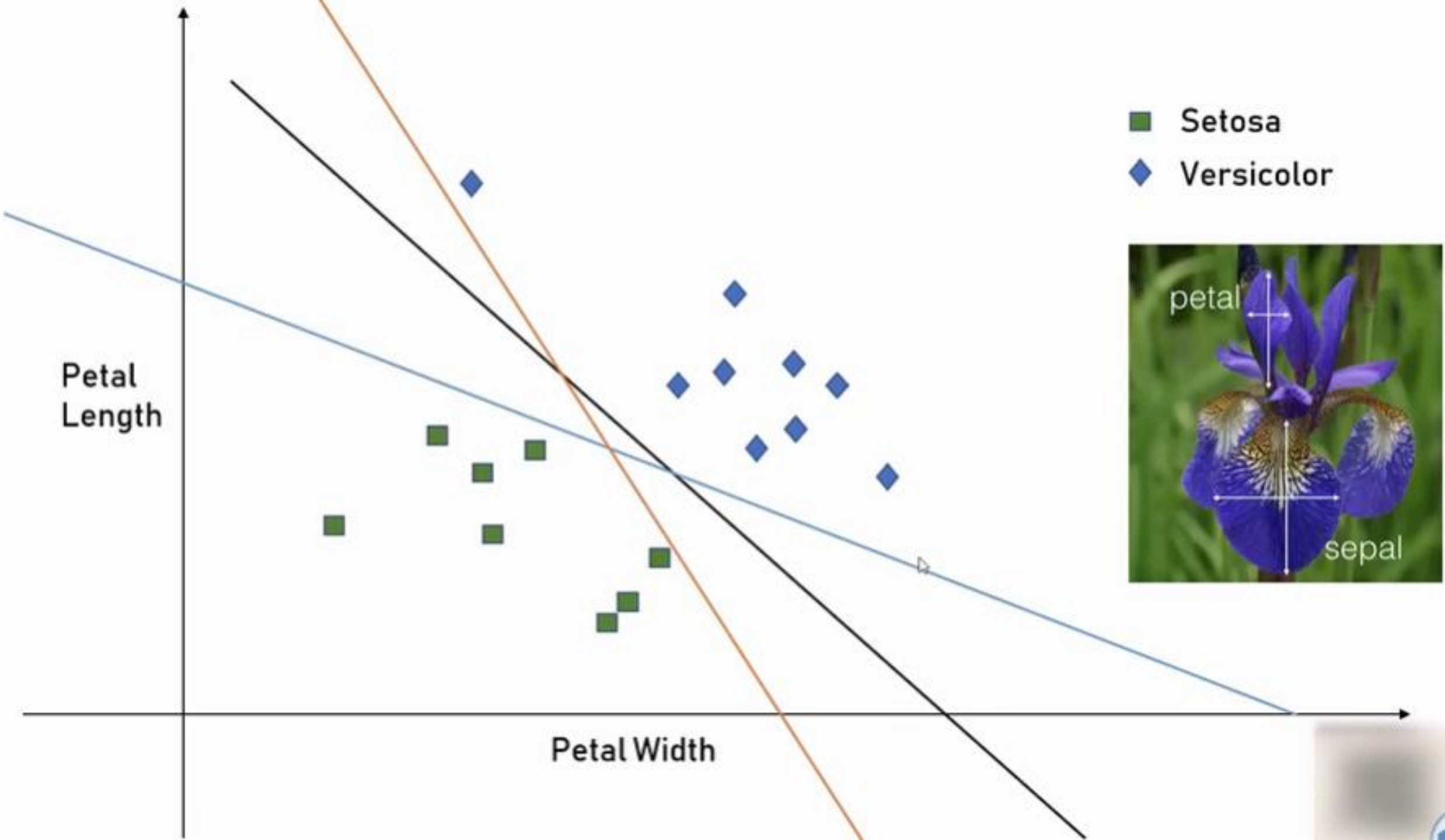
Overview

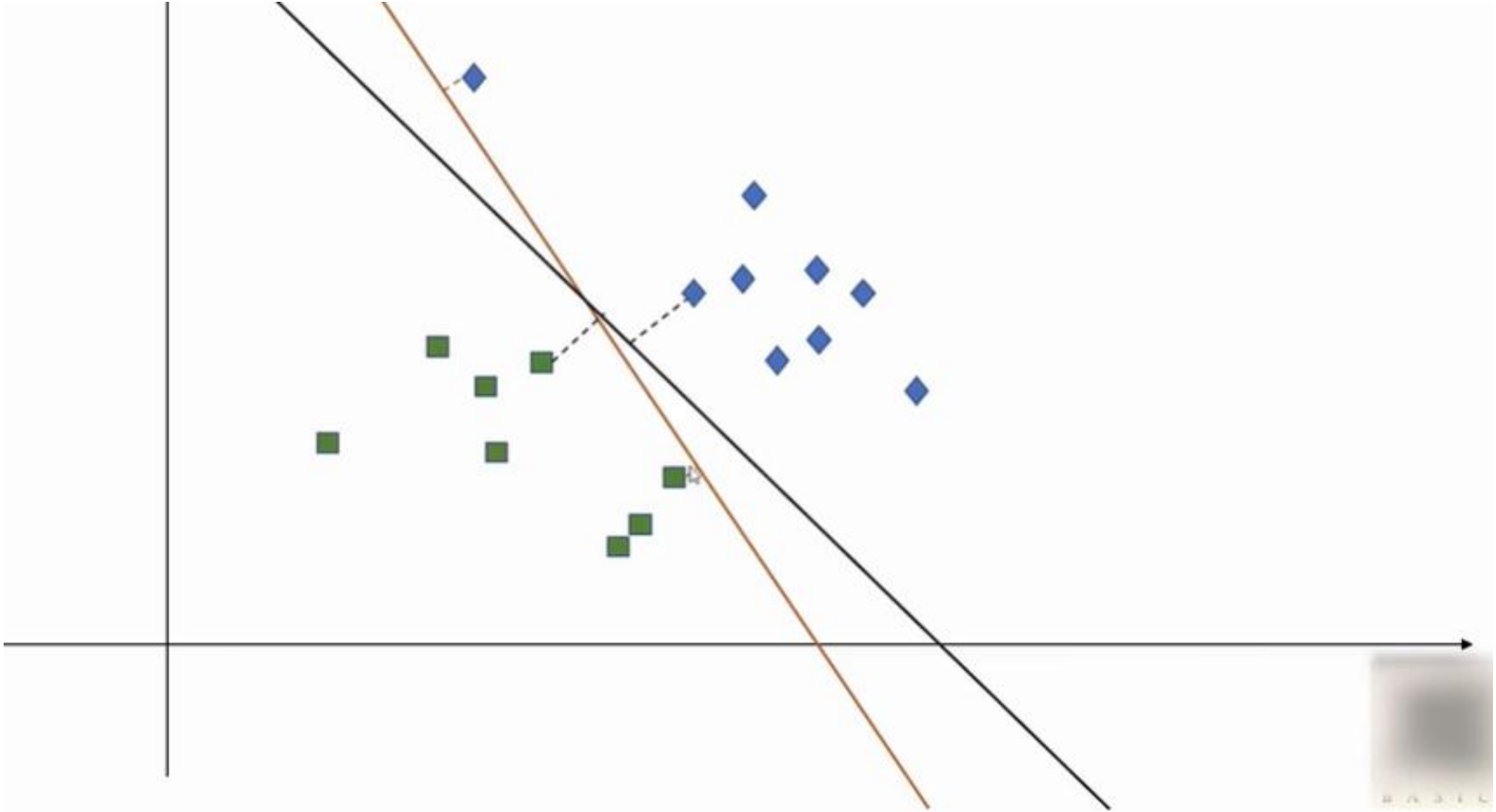
- Classification algorithms
  - Support Vector Machine
  - Decision Tree
  - Random Forest

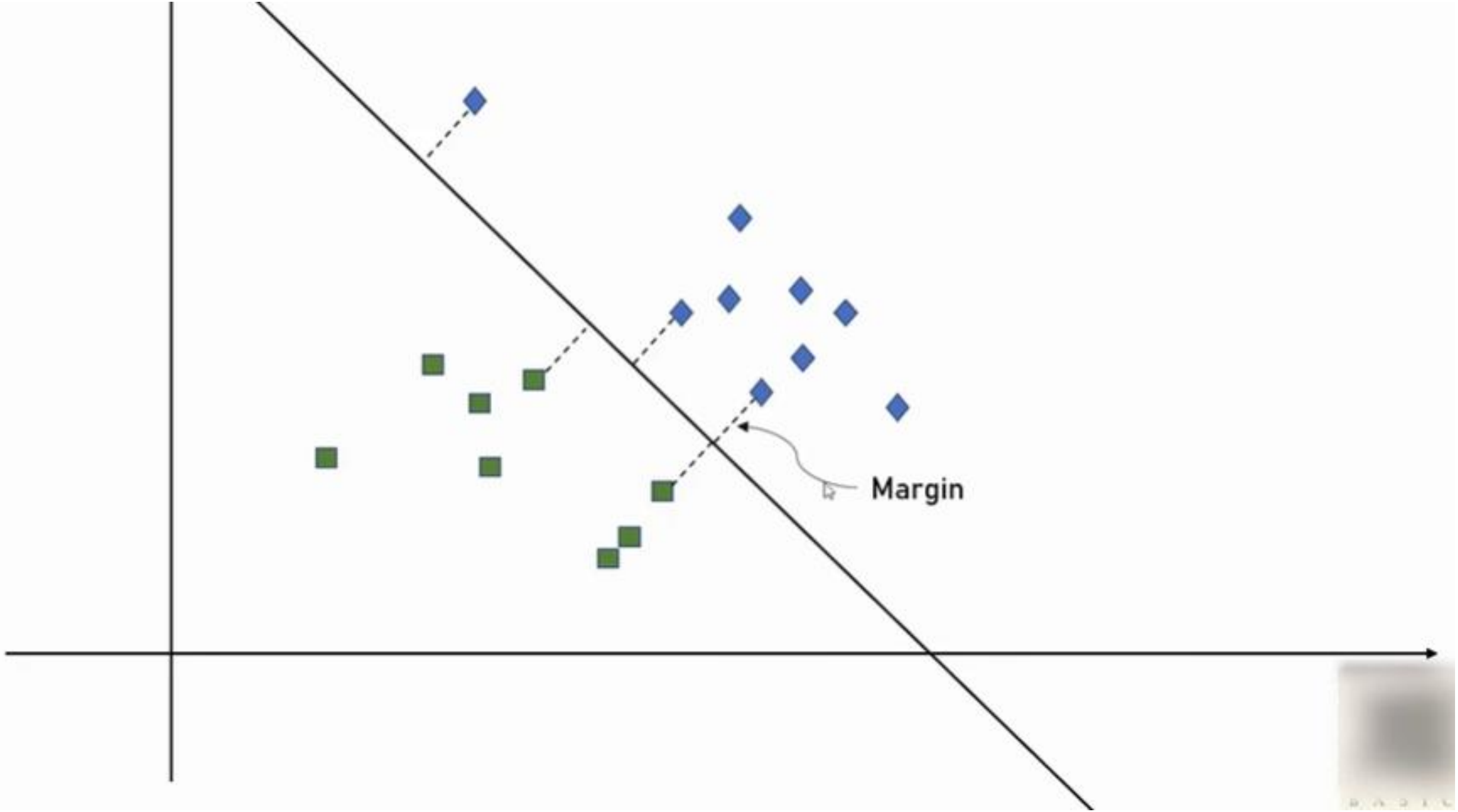
Support Vector Machine (SVM)

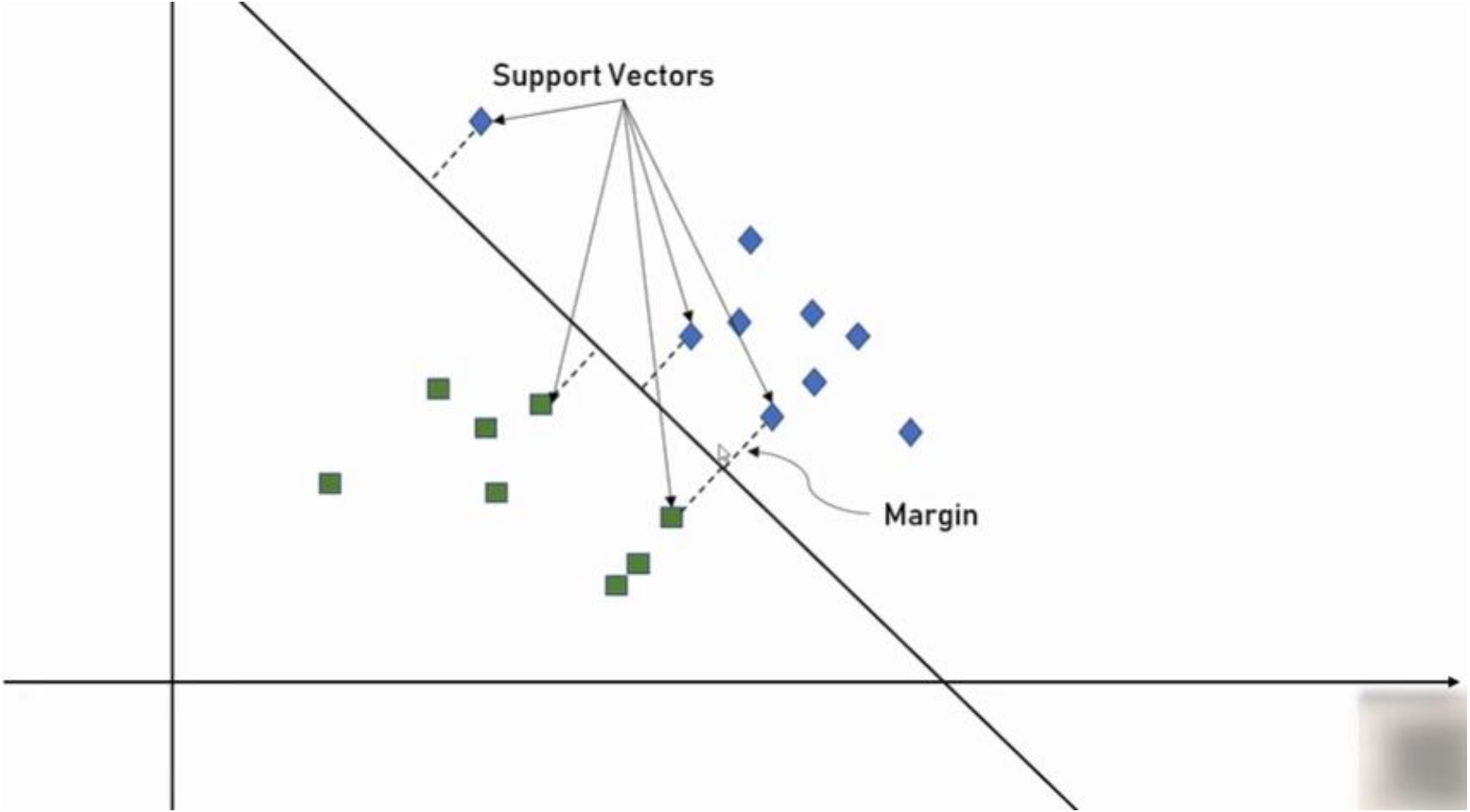
Classification


Setosa
Versicolor
petal
sepal
Petal Length
Petal Width

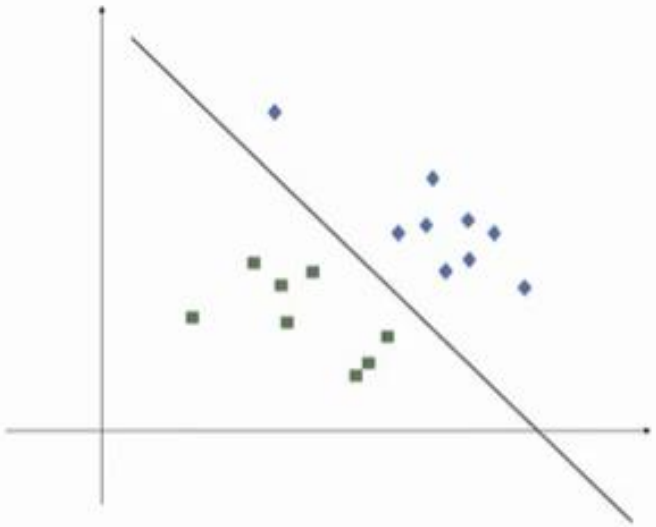

Setosa
Versicolor
Petal Length
Petal Width
petal
sepal




Margin


Support Vectors
Margin

2D



3D

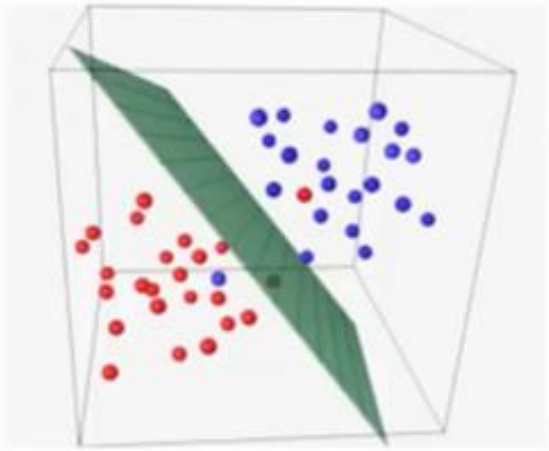
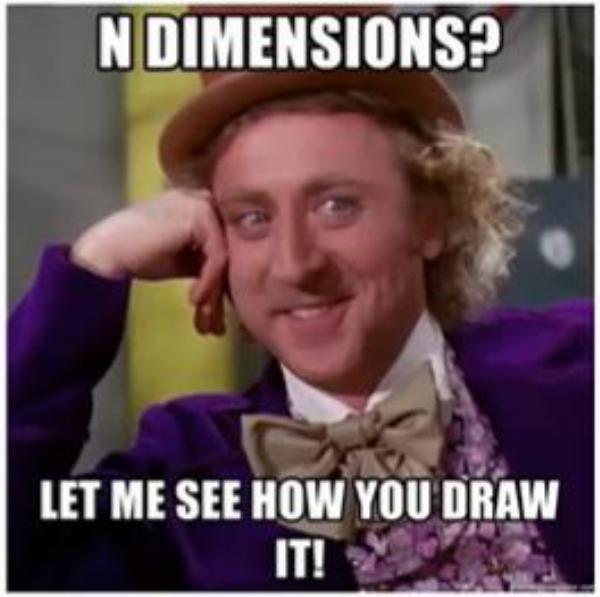


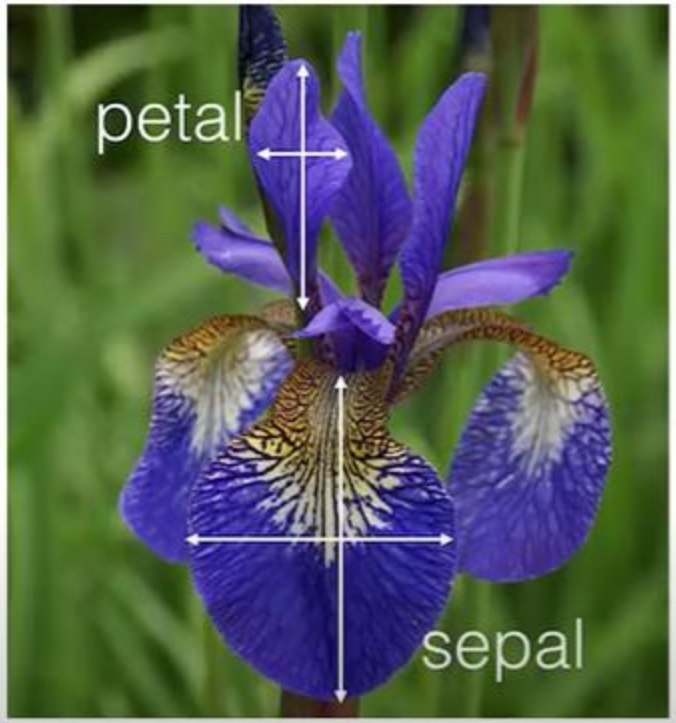
Image Credit: https://appliedmachinelearning.blog

nD



Support vector machine draws a hyper plane in n dimensional space such that it maximizes margin between classification groups

sklearn iris flower dataset



features

target label

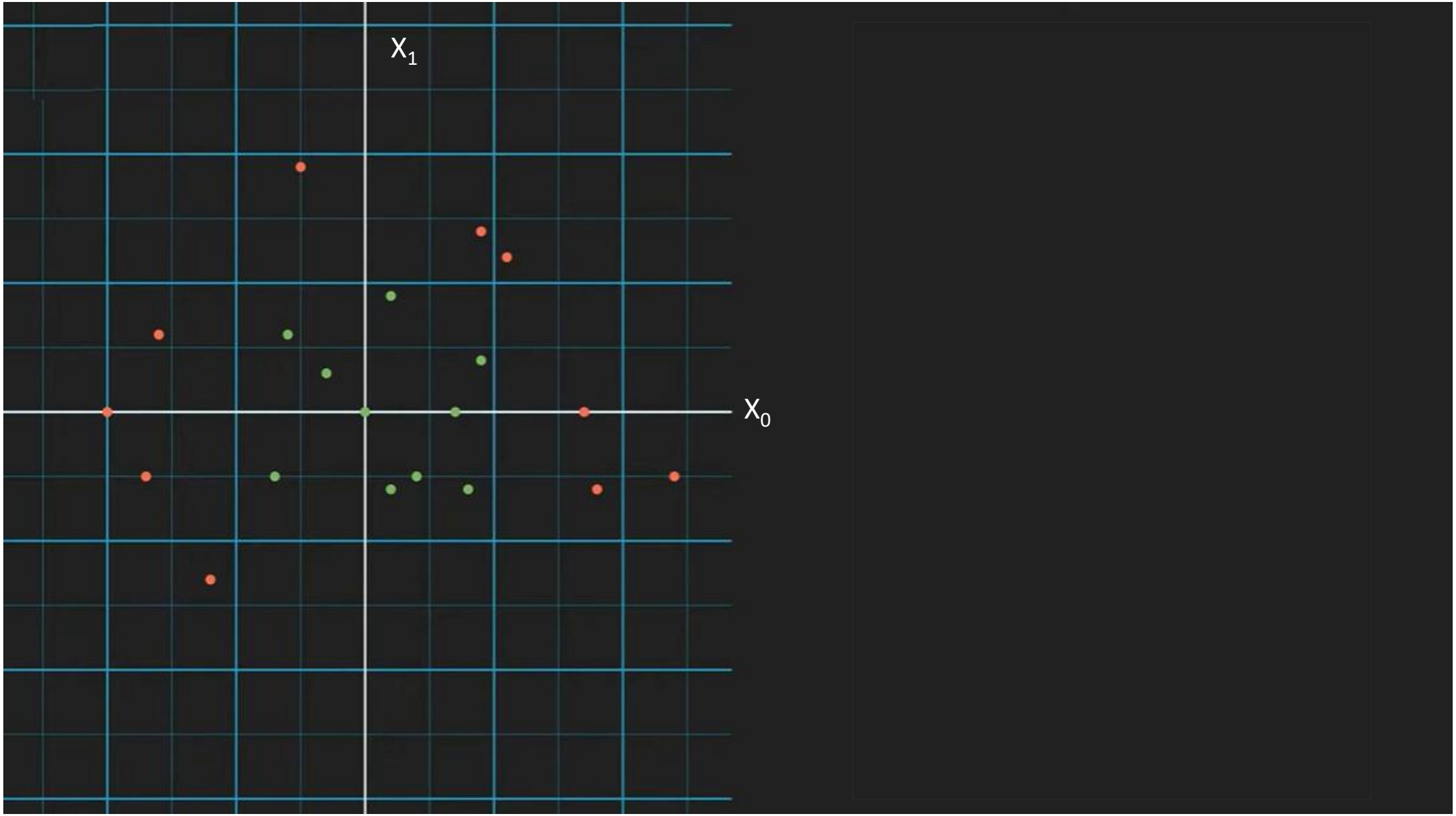
sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	flower
4.6	3.2	1.4	0.2	setosa
5.3	3.7	1.5	0.2	setosa
5.0	3.3	1.4	0.2	setosa
7.0	3.2	4.7	1.4	versicolor
6.4	3.2	4.5	1.5	versicolor

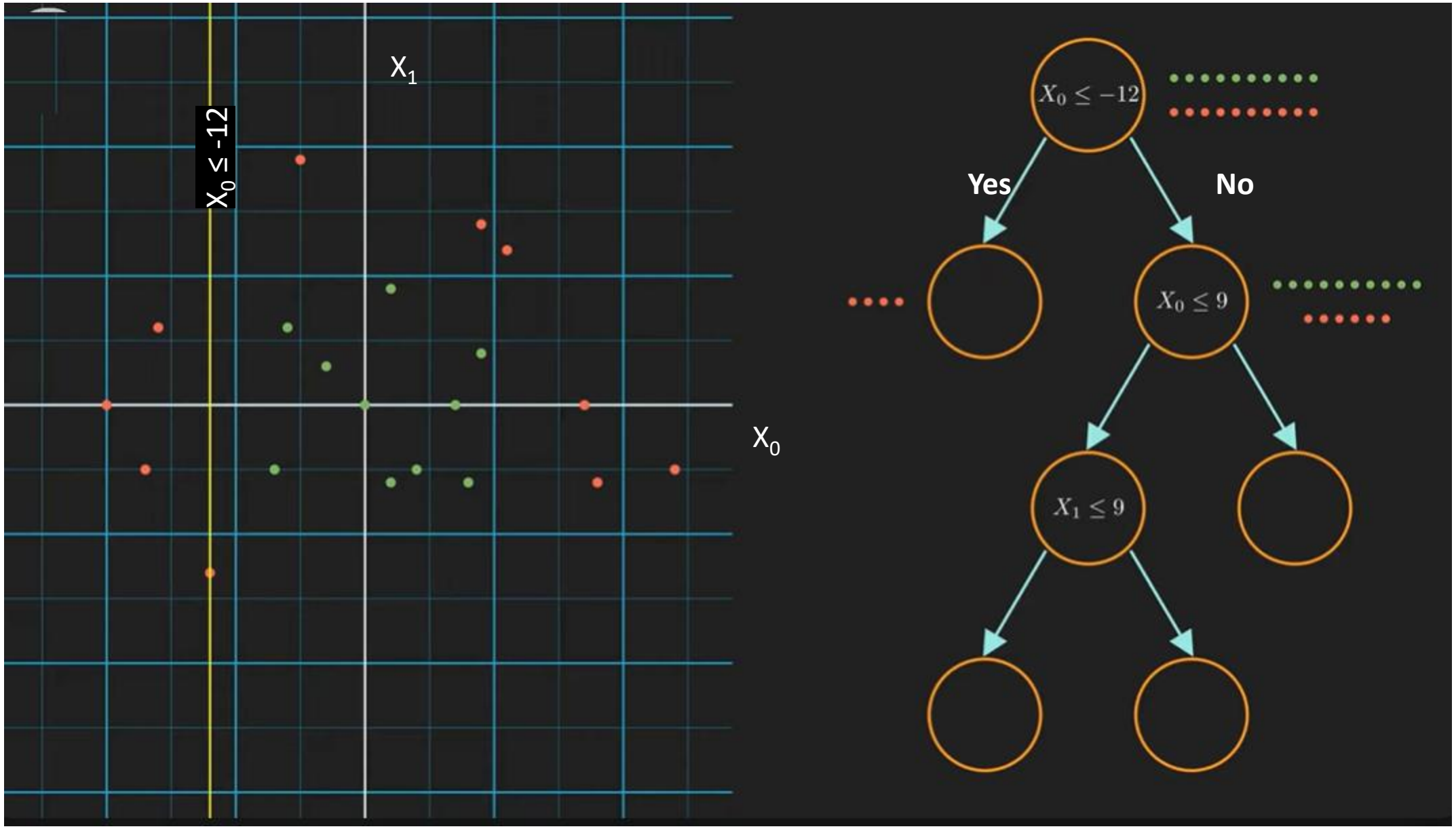
Demo

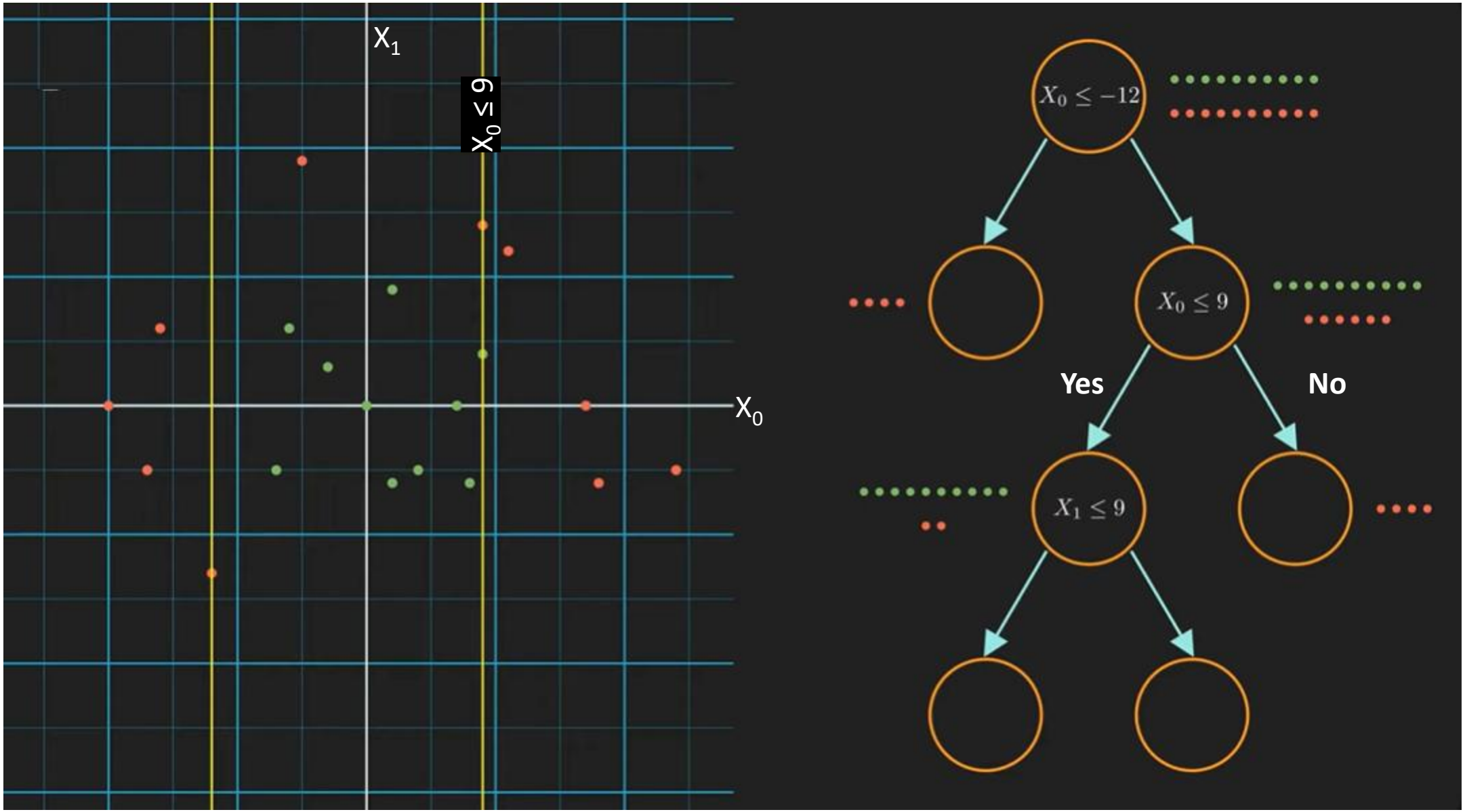
SVM

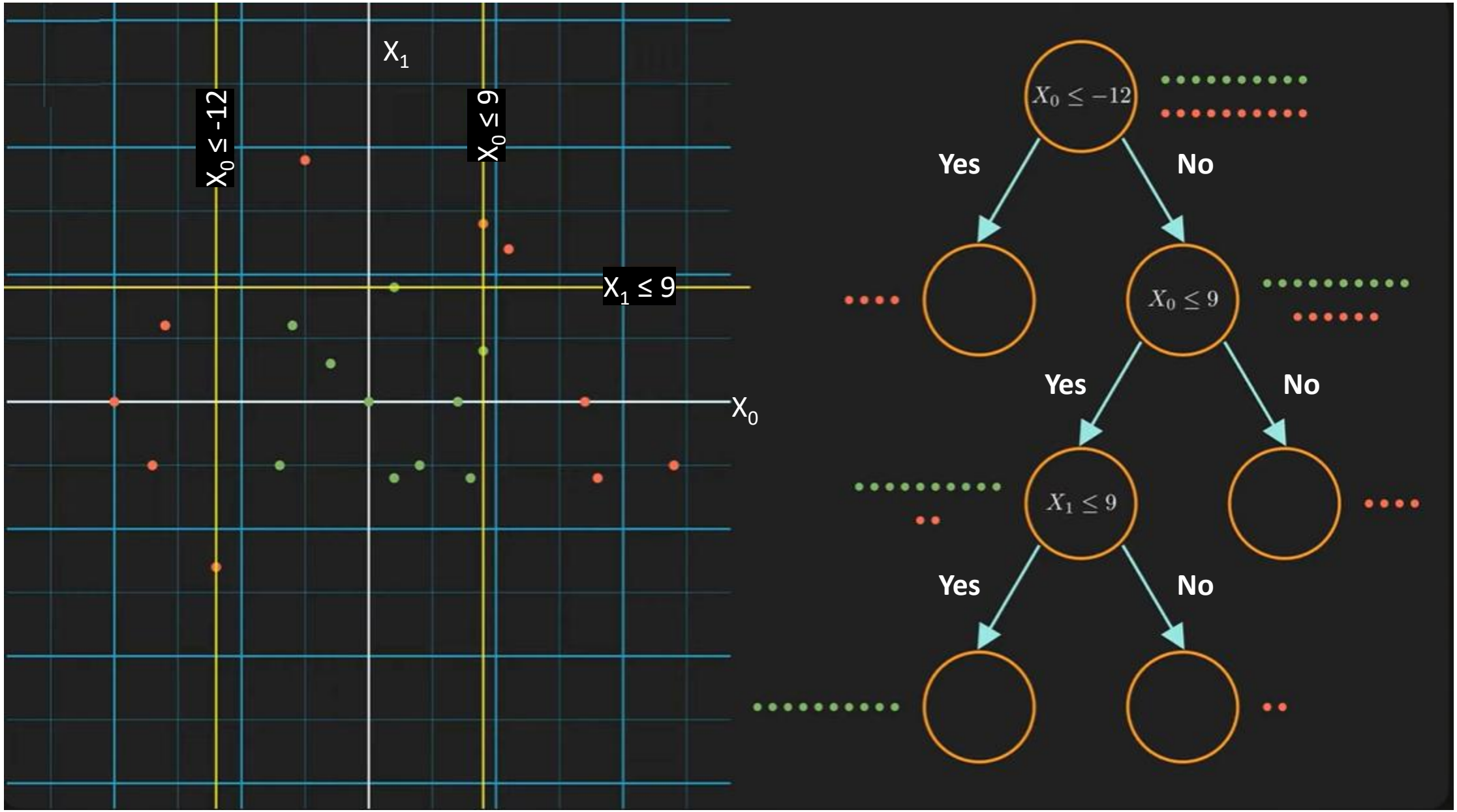
Decision Tree

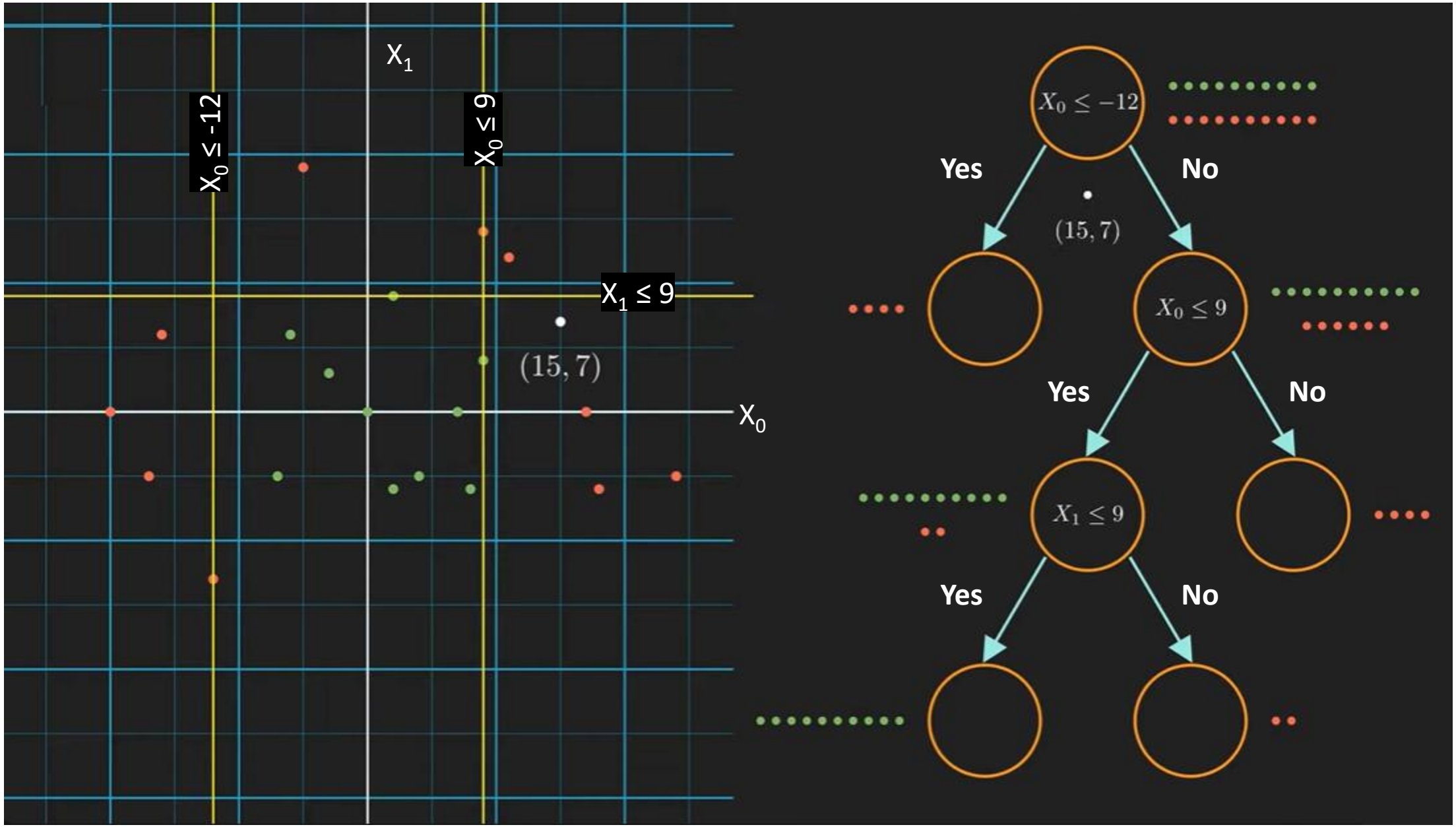
Classification

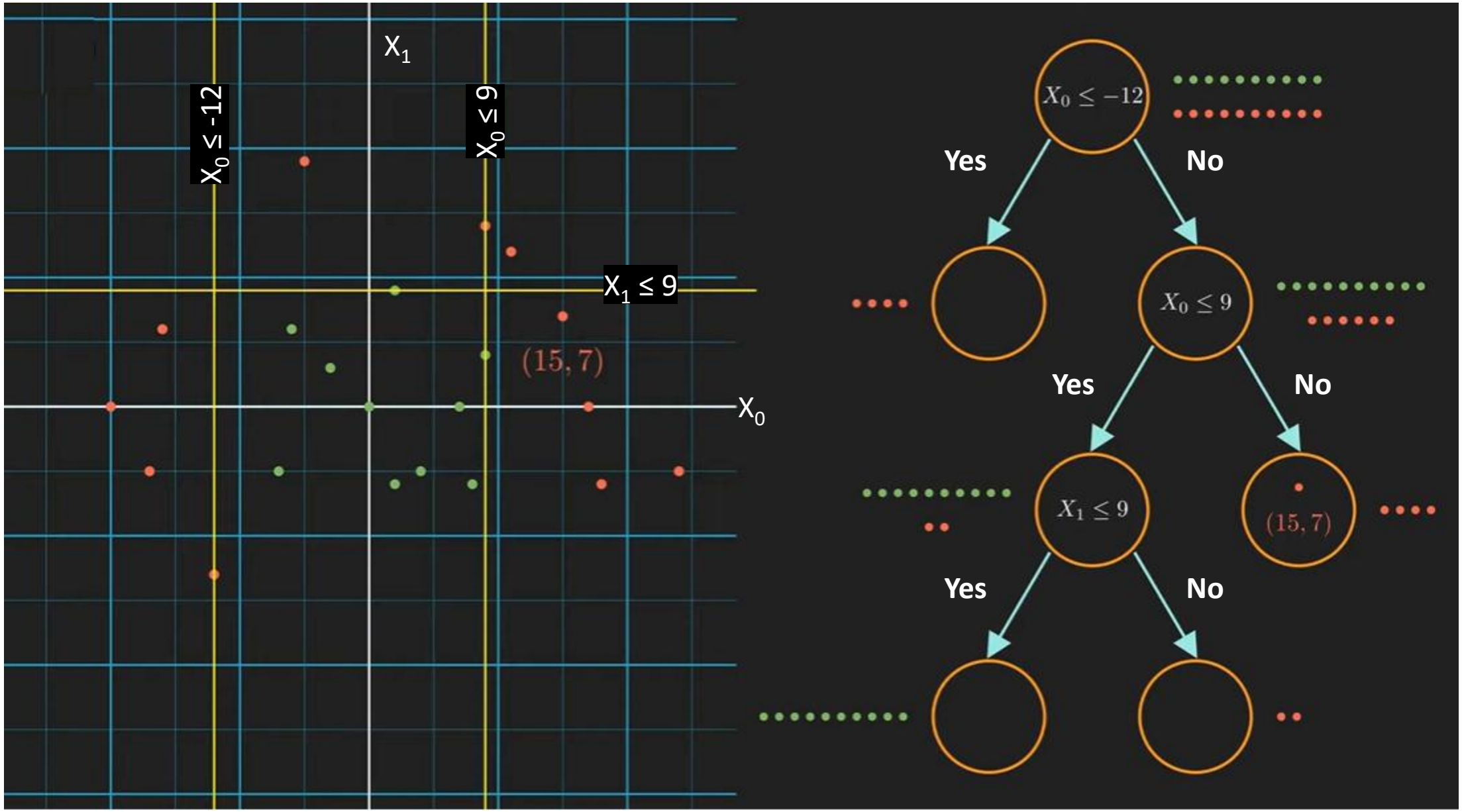

x_1
x_0

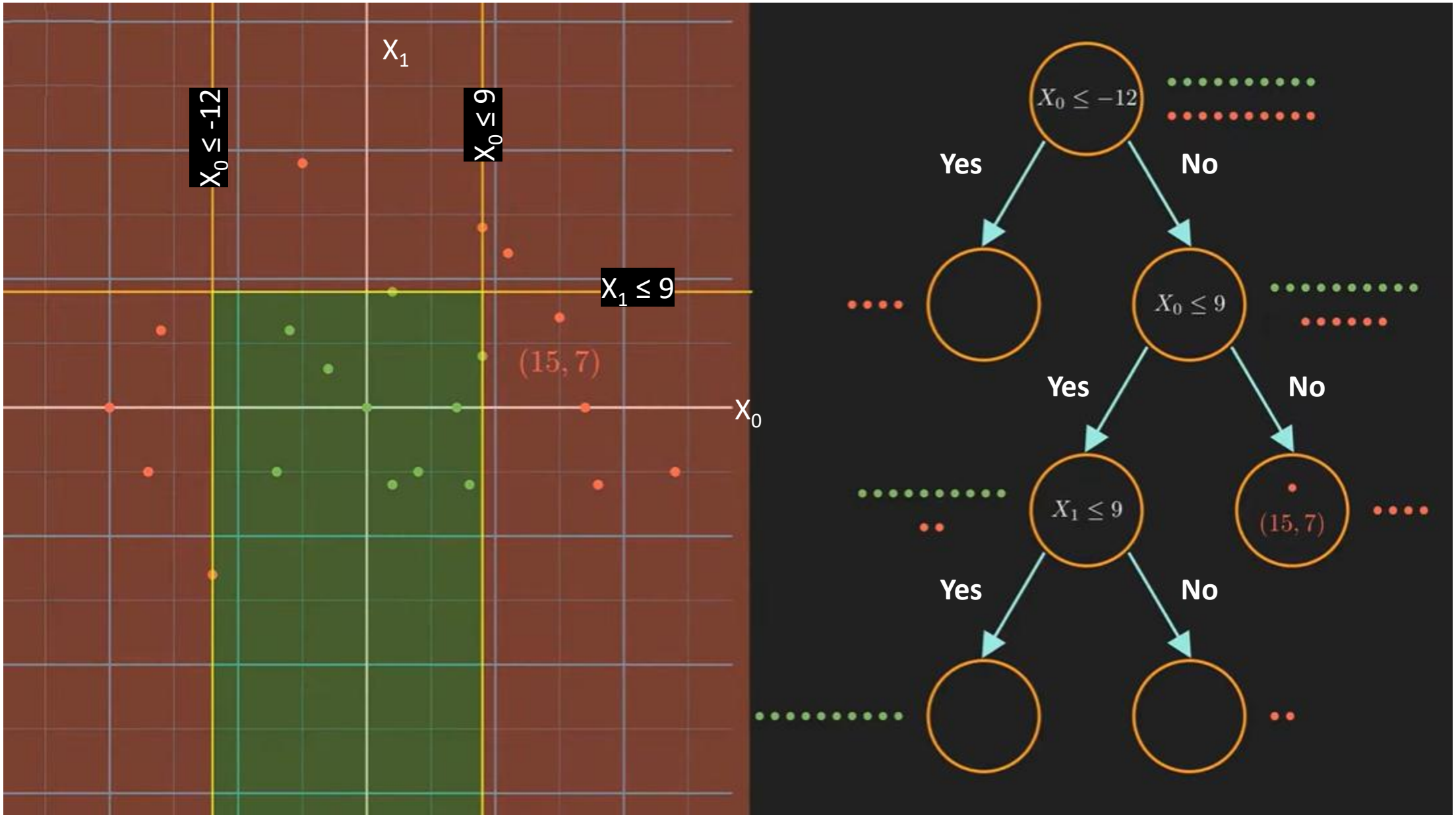

X_1
$X_0 \leq -12$
X_0
$X_0 \leq -12$
Yes
No
$X_0 \leq 9$
$X_1 \leq 9$

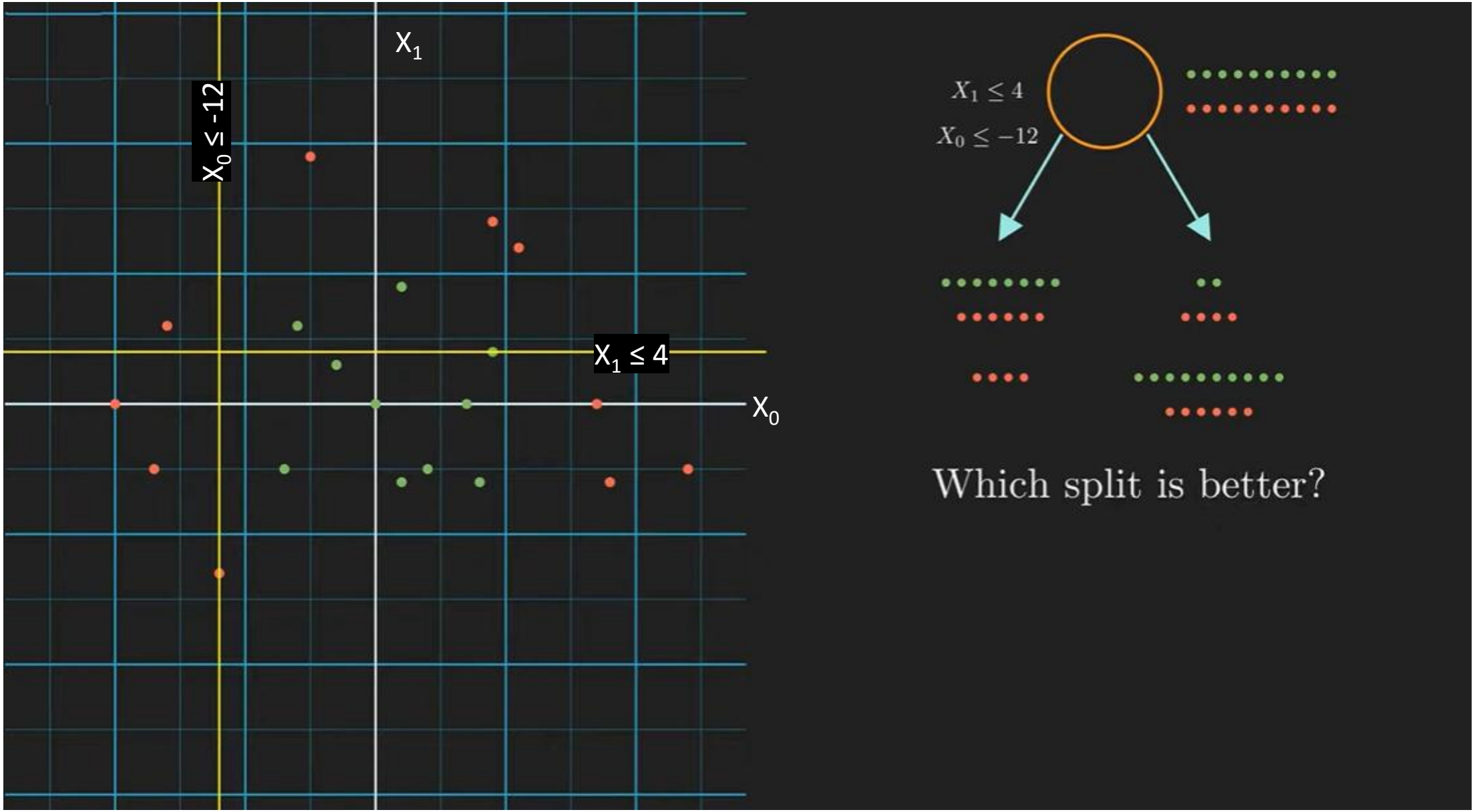

X_1
$X_0 \leq 9$
X_0
$X_0 \leq -12$
$X_0 \leq 9$
Yes
No
$X_1 \leq 9$

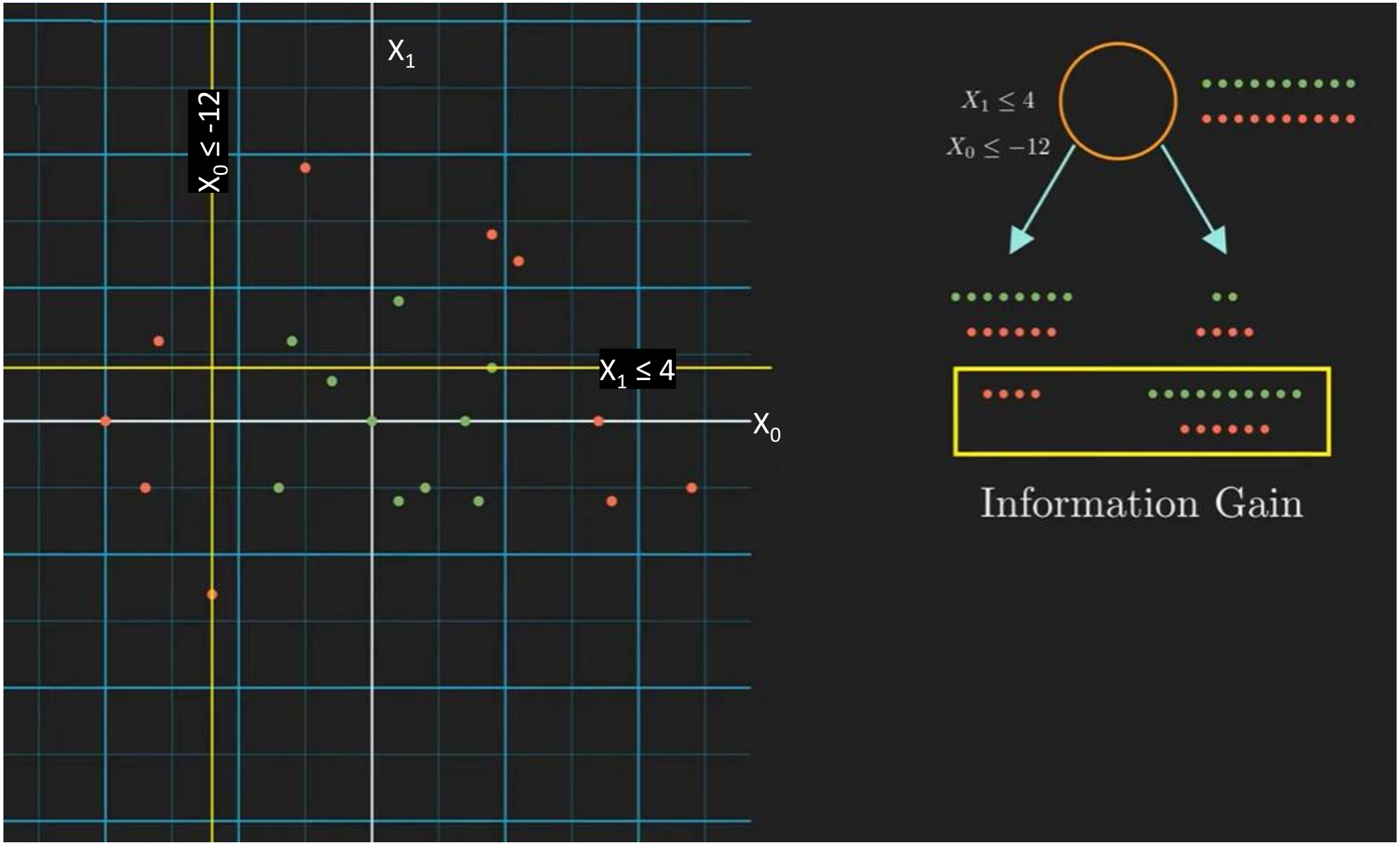

X_1
$X_0 \leq -12$
$X_0 \leq 9$
$X_1 \leq 9$
X_0
$X_0 \leq -12$
Yes
No
$X_0 \leq 9$
Yes
No
$X_1 \leq 9$
Yes
No

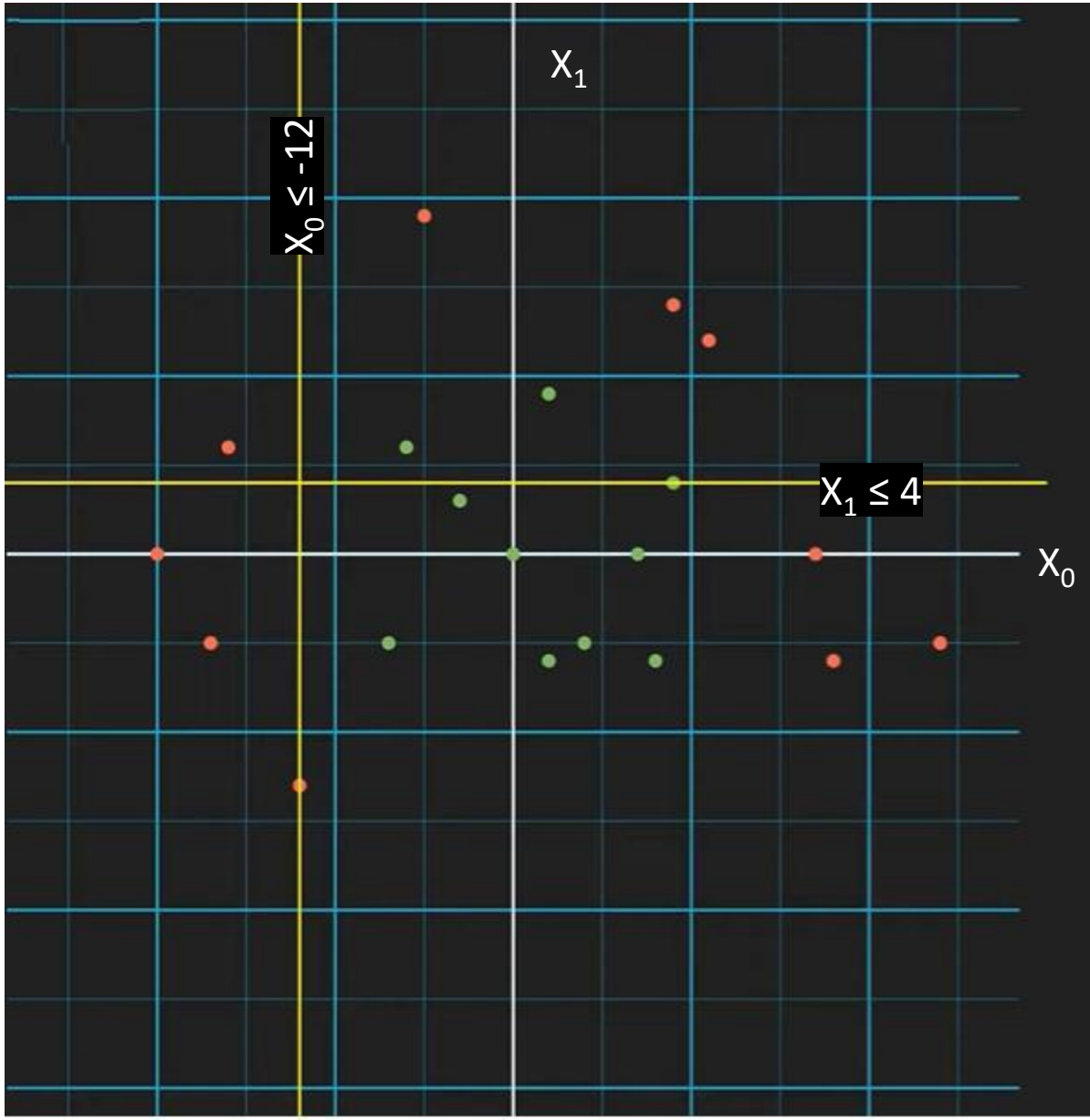

X_1
$X_0 \leq -12$
$X_0 \leq 9$
$X_1 \leq 9$
(15, 7)
X_0
$X_0 \leq -12$
Yes
No
(15, 7)
$X_0 \leq 9$
Yes
No
$X_1 \leq 9$
Yes
No

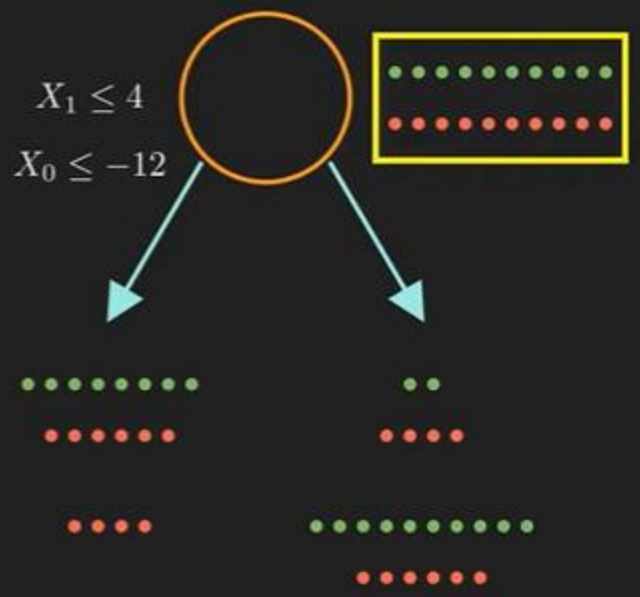

X_1
$X_0 \leq -12$
$X_0 \leq 9$
$X_1 \leq 9$
(15, 7)
X_0
$X_0 \leq -12$
Yes
No
$X_0 \leq 9$
Yes
No
$X_1 \leq 9$
(15, 7)
Yes
No


X_1
$X_0 \leq -12$
$X_0 \leq 9$
$X_1 \leq 9$
(15, 7)
X_0
$X_0 \leq -12$
Yes
No
$X_0 \leq 9$
Yes
No
$X_1 \leq 9$
(15, 7)
Yes
No


X_1
$X_0 \leq -12$
$X_1 \leq 4$
X_0
$X_1 \leq 4$
$X_0 \leq -12$
Which split is better?


X_1
$X_0 \leq -12$
$X_1 \leq 4$
X_0
$X_1 \leq 4$
$X_0 \leq -12$
Information Gain

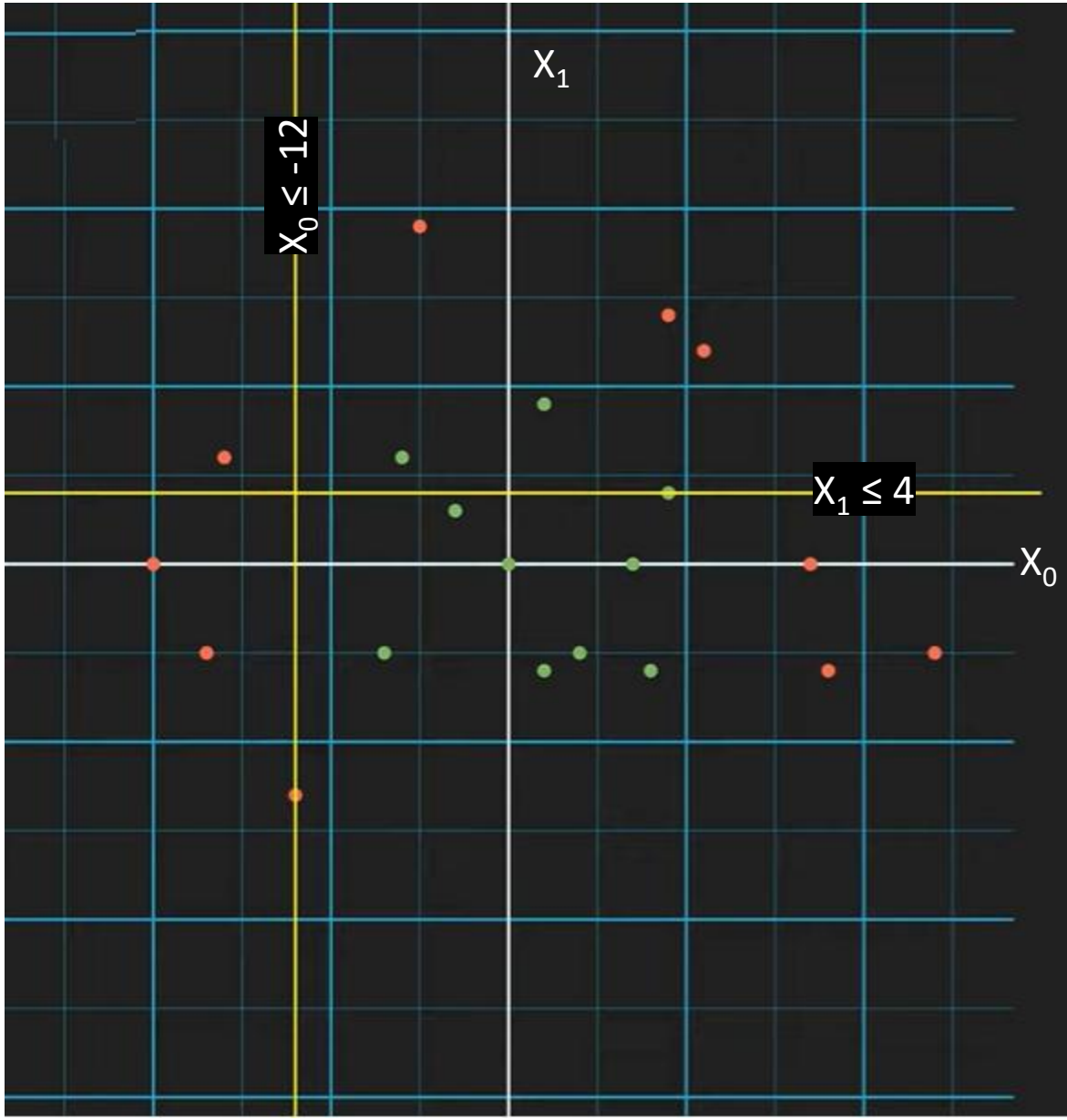


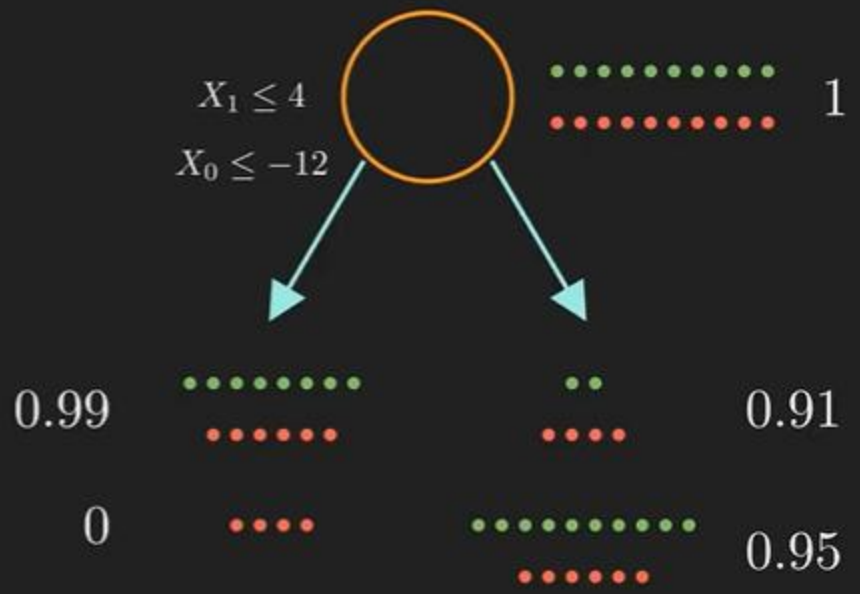


$$Entropy = \sum - p_i \ \log(p_i)$$

p_i = probability of class i

$$-0.5\log(0.5) - 0.5\log(0.5)$$

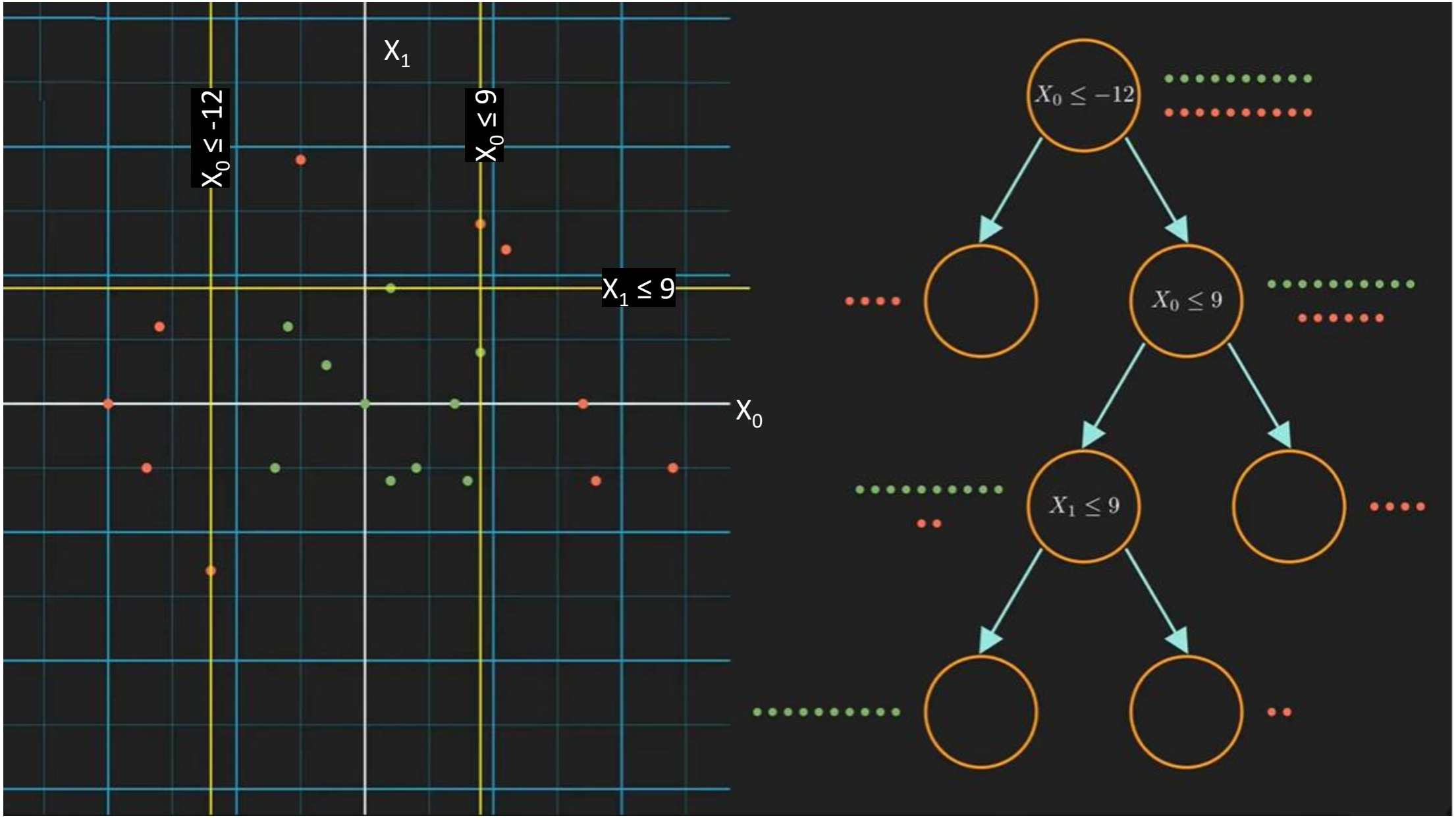




$$\mathrm{IG} = E(parent) - \sum w_i \; E(child_i)$$

$$\mathrm{IG}_1 = 1 - \frac{14}{20} \times .99 - \frac{6}{20} \times .91 = 0.034$$

$$\mathrm{IG}_2 = 1 - \frac{4}{20} \times 0 - \frac{16}{20} \times .95 = 0.24$$


X_1
$X_0 \leq -12$
$X_0 \leq 9$
$X_1 \leq 9$
X_0
$X_0 \leq -12$
$X_0 \leq 9$
$X_1 \leq 9$

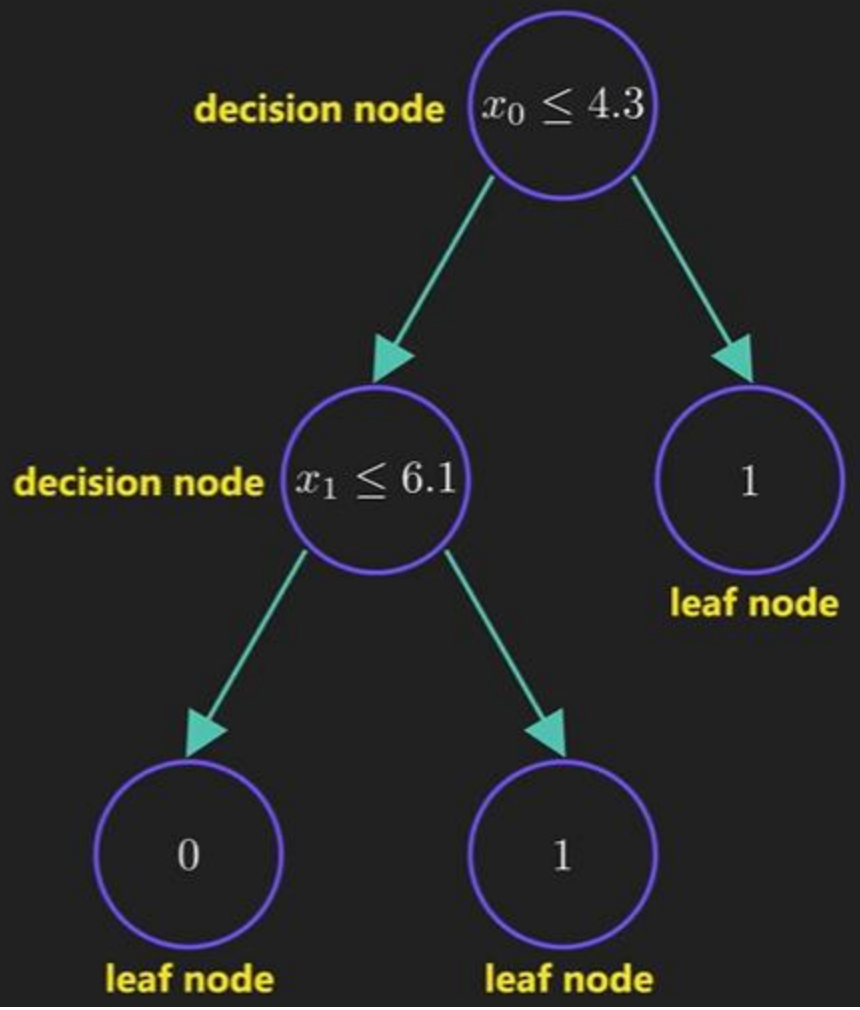
Demo

Decision Tree

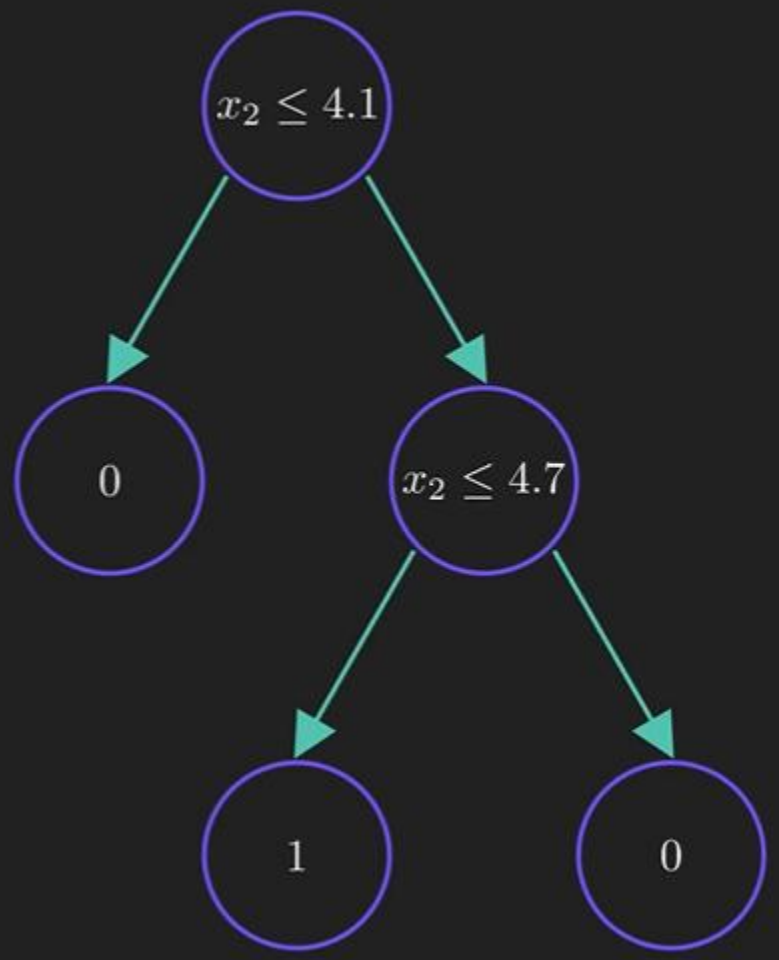
Random Forest

Classification

id	x_0	x_1	x_2	x_3	x_4	y
0	4.3	4.9	4.1	4.7	5.5	0
1	3.9	6.1	5.9	5.5	5.9	0
2	2.7	4.8	4.1	5.0	5.6	0
3	6.6	4.4	4.5	3.9	5.9	1
4	6.5	2.9	4.7	4.6	6.1	1
5	2.7	6.7	4.2	5.3	4.8	1



id	x_0	x_1	x_2	x_3	x_4	y
0	4.3	4.9	4.1	4.7	5.5	0
1	6.5	4.1	5.9	5.5	5.9	0
2	2.7	4.8	4.1	5.0	5.6	0
3	6.6	4.4	4.5	3.9	5.9	1
4	6.5	2.9	4.7	4.6	6.1	1
5	2.7	6.7	4.2	5.3	4.8	1



id	x_0	x_1	x_2	x_3	x_4	y
0	4.3	4.9	4.1	4.7	5.5	0
1	3.9	6.1	5.9	5.5	5.9	0
2	2.7	4.8	4.1	5.0	5.6	0
3	6.6	4.4	4.5	3.9	5.9	1
4	6.5	2.9	4.7	4.6	6.1	1
5	2.7	6.7	4.2	5.3	4.8	1

id	id	id	id
2	2	4	3
0	1	1	3
2	3	3	2
4	1	0	5
5	4	0	1
5	4	2	2

Bootstrapped Datasets

id	x_0	x_1	x_2	x_3	x_4	y
0	4.3	4.9	4.1	4.7	5.5	0
1	3.9	6.1	5.9	5.5	5.9	0
2	2.7	4.8	4.1	5.0	5.6	0
3	6.6	4.4	4.5	3.9	5.9	1
4	6.5	2.9	4.7	4.6	6.1	1
5	2.7	6.7	4.2	5.3	4.8	1

id
2
0
2
4
5
5

x_0, x_1

id
2
1
3
1
4
4

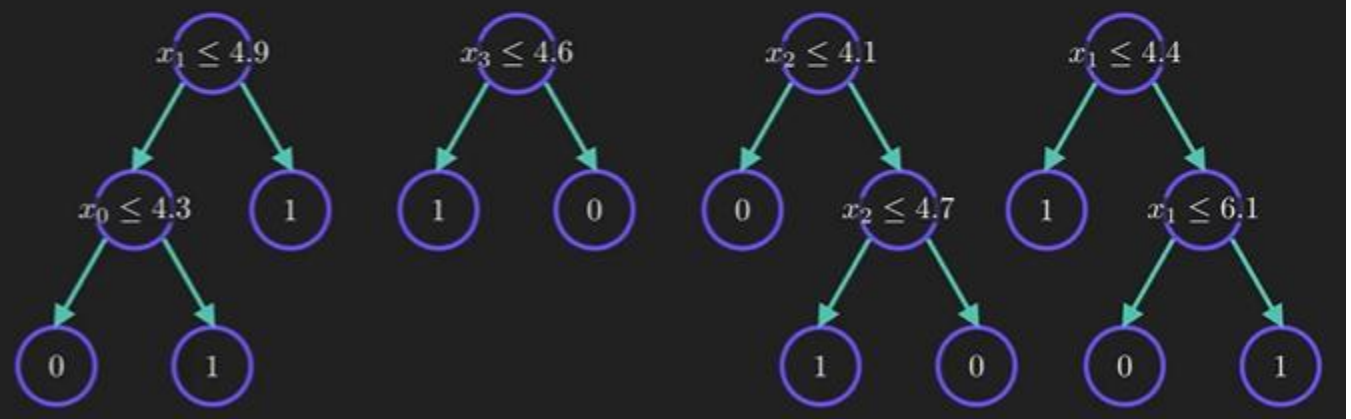
x_2, x_3

id
4
1
3
0
0
2

x_2, x_4

id
3
3
2
5
1
2

x_1, x_3



id	x_0	x_1	x_2	x_3	x_4	y
0	4.3	4.9	4.1	4.7	5.5	0
1	3.9	6.1	5.9	5.5	5.9	0
2	2.7	4.8	4.1	5.0	5.6	0
3	6.6	4.4	4.5	3.9	5.9	1
4	6.5	2.9	4.7	4.6	6.1	1
5	2.7	6.7	4.2	5.3	4.8	1

id
2
0
2
4
5
5

x_0, x_1

id
2
1
3
1
4
4

x_2, x_3

id
4
1
3
0
0
2

x_2, x_4

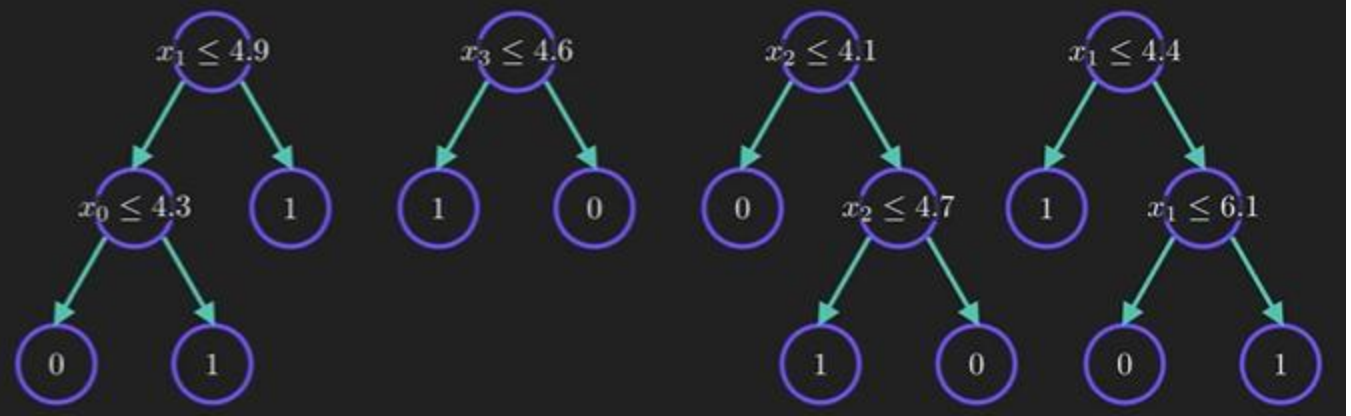
id
3
3
2
5
1
2

x_1, x_3

2.8	6.2	4.3	5.3	5.5

Bootstrap + Aggregating

(Bagging)



1 0 1 1

Why is it called random?

Why Bootstrapping and Feature Selection?

How many features to consider?

How to use this for regression?

Demo

Random Forest